Roll No.

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the course: B. Tech
Semester: 3rd
Name of Paper: Production Technology
Paper code: TME 309
Time: 3 Hours
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Describe in detail the types of patterns and allowances used in foundry practice. Illustrate with neat sketches.	CO1, CO2
(b)	Explain the design of gating and risering systems in casting. Derive the relationships used in riser volume and feeding distance calculations.	
(c)	What are the types and properties of molding sands? How are molding sand properties tested and controlled in foundry operations?	
Q2	(20 marks)	
(a)	Discuss yield criteria for ductile materials such as Tresca and Von Mises. Compare their applications in metal forming analysis.	
(b)	Derive the flow curve equation ($\sigma = K\varepsilon^n$) for plastic deformation and explain its significance in metal forming.	CO1, CO3
(c)	Explain the mechanics of metalworking processes and describe the classification of these processes based on stress conditions.	
Q3	(20 marks)	
(a)	Explain the classification of rolling processes and describe the construction and working of various rolling mills.	CO3, CO5
(b)	Derive an expression for rolling load and roll separating force, and explain the influence of friction, tension, and reduction ratio.	
(c)	Discuss the defects in rolled products and describe the methods to minimize them. How does frictional loss affect rolling power requirements?	
Q4	(20 marks)	
(a)	Compare TIG and MIG welding processes with respect to principle, equipment, and applications.	C03, CO4
(b)	Explain the process and equipment used in gas welding and gas cutting. Discuss the types of flames and their specific uses.	
(c)	Describe resistance welding, submerged arc welding, and friction welding processes. Differentiate them from soldering and brazing with examples.	
Q5	(20 marks)	
(a)	Explain welding distortion and its types. What are the methods of reducing residual stresses and distortion in welded structures?	C05, CO6
(b)	Discuss the concept of weldability of materials. How is weldability tested and evaluated for ferrous and non-ferrous metals?	
(c)	Explain the defects in welding, their causes and preventive measures, and their influence on the performance of welded components.	